P.6 Analyzing Graphs of Functions

Name _____ Period ____ Date _____

Pre-Calculus P.6 · Student sheet

I can…

- use the Vertical Line Test and find the zeros of a function.
- find where a function increases or decreases and find its average rate of change.
- decide whether a function is even, odd, or neither.

Vocabulary · fill in the missing word

Zeros of a function — the _____ where $f(x) = 0$

Average rate of change — $\dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$, the slope of the _____ line

Even and odd functions — even: $f(-x) =$ _____; odd: $f(-x) = -f(x)$

Part 1 · Read the graph

Graph $f(x) = x^3 - 3x$.

Zeros	Increasing on	Decreasing on	Turning points	f(2) and f(−2)

Part 2 · Notes and examples (fill in each note, then work the example)

Vertical Line Test: a graph is a function if no vertical line meets it more than _____.

Example 1

Is each graph a function?

a. $x = y^2$

b. $y = |x|$

Zeros: solve $f(x) =$ _____; they are the x-intercepts.

Example 2

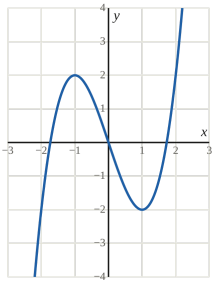
Find the zeros.

a. $f(x) = 3x^2 + x - 10$

b. $g(x) = \sqrt{10 - x^2}$

Example 3

Use the graph of $f(x) = x^3 - 3x$. Where is f increasing and decreasing? Find the relative extrema.



Average rate of change from x_1 to x_2: _____

Example 4

Find the average rate of change of $f(x) = x^3 - 3x$ from (a) $x = -2$ to 0 and (b) $x = 0$ to 1.

Even: $f(-x) = f(x)$, symmetric about the y-axis. **Odd:** $f(-x) =$ _____, symmetric about the origin.

Example 5

Even, odd, or neither?

a. $g(x) = x^3 - x$

b. $h(x) = x^2 + 1$

c. $f(x) = x^3 - 1$

Part 3 · Partner practice

Partner 1

Find the zeros of $f(x) = x^2 - 5x - 14$.

Partner 2

Find the average rate of change of $f(x) = x^2 + 1$ from $x = 1$ to $x = 3$.